

EduGenie: Google Gemini Powered Learning Assistant

2. Requirement Analysis Phase

Purpose: Determine the functional, technical, and setup requirements needed to build EduGenie.

Phase Activities

- Functional requirements: Q&A;, simplified explanations, quiz generation, summarization, and personalized learning recommendations.
- Python 3.10+ is required.
- FastAPI is used for the backend.
- HTML and CSS are used for the web interface.
- A Google Gemini API key is required for cloud-based AI functions.
- Uvicorn is used as the ASGI server and Jinja2 is used for HTML templating.
- Gemini 1.5 Pro is used for Q&A;, summarization, quiz generation, and learning paths; LaMini-Flan-T5-783M is used for concept explanation.

Project Focus: This phase is organized from the supplied EduGenie project document while retaining the requested phase name.